Deep Bhanderi

+1-352-870-5951 | bhanderija@gmail.com | LinkedIn | Gainesville, Florida

EDUCATION

University of Florida | GPA: 3.95/4.0 **Aug'25 – May'27**
Master of Science (MS), Electrical and Computer Engineering
Coursework: Advanced VLSI I & II, Reconfigurable Computing, Adv. Verification with UVM, Computer Architecture

The Maharaja Sayajirao University, Baroda | GPA: 3.85/4.0 **Aug'19 – Jun'23**
Bachelor of Engineering, Electronics Engineering
Coursework: Digital Logic Design, Analog Electronics, Microprocessors, VLSI Design, Control Systems

SKILLS

Programming & HDL: C++, Linux, VHDL, Verilog, System Verilog, Python.
Verification: UVM Testbench development, Assertion-Based Verification (SVA), CRV, Verification Plans
EDA Tools: Xilinx Vivado, QuestaSim, Quartus Prime, Synopsys VCS, Cadence Virtuoso, gem5.
Concepts: RTL Design, Computer Architecture, Functional Verification, Timing Optimization, RISC-V, FPGA prototyping.
Bus Protocols: I2C, SPI, UART, AMBA AXI4-Stream.

EXPERIENCE

Graduate Teaching Assistant **Aug'26 – Present**
Computer Architecture (EEL 5764) *University of Florida*
- Support **Dr. Sandip Ray** in delivering graduate **Computer Architecture coursework** (superscalar pipelines, memory hierarchies, performance analysis) to 80 students, covering simulation environments, homework design, and grading.

SHREC Research Centre **Apr'26 – Present**
Research Assistant *University of Florida*
- Research in **High-Performance Computing (HPC) under Dr. Herman Lam**, with a focus on FPGA-based kernel acceleration, optimizing AXI4-integrated RTL kernels on a Xilinx Arty A7-35T.
- Building CPU microbenchmarks as baseline for comparing kernel throughput against Tenstorrent n300 accelerator.

GNFC (State Govt. Company) **Jun'23 – May'25**
Electronics Lab Engineer *Bharuch, Gujarat, India*
- Diagnosed and validated digital and **mixed-signal circuits** (Digital I/O, Amplifier system cards) using signal generators, oscilloscope for root cause analysis, reducing maintenance costs by **~USD 10,000 annually.**
- Performed **IC-level fault diagnosis on PCBs** using stuck-at fault isolation, continuity checks, and signal probing applying fault modeling concepts directly analogous to DFT.

PROJECTS

BNN Hardware Accelerator for digit recognition on FPGA (apple project) | Design optimization, sv **In progress**
- Implementing a BNN classifier accelerator on Xilinx Ultra Scale+ FPGA using **XNOR-pop count**-threshold neuron processing for area-efficient inference across a 784→256→256→10 topology with parameterized RTL.

MAC Datapath Timing Closure via Pipelining **|** SystemVerilog, Quartus, Questa
- Increased restricted **Fmax 2.4×** (104 MHz to 250 MHz) on a pipelined multiply-accumulate datapath by iteratively extracting critical paths in Quartus Timing Analyzer.
- Verified functional equivalence in Questa, keeping 5-cycle added latency within 5% of the FSMD's 32-cycle interval.

Out-of-Order Superscalar Processor using Tomasulo's Algorithm **|** System Verilog, ModelSim, RTL Design
- Designed a Tomasulo-based OoO processor with reservation stations, register renaming, reorder buffers, and CDB arbitration for **dynamic scheduling and hazard resolution**.
- Built self-checking testbenches in ModelSim validating dependency handling, commit ordering, and pipeline correctness.

AXI4-Stream Networking Packet Filter Verification | UVM Verification, System Verilog
- Designed a reusable layered test bench and verified a streaming packet filter using the AXI4-Stream protocol with full UVM testbench infrastructure incorporating Covergroups, **Cross-coverage, and SV assertions** to detect bugs.
- Extended DUT with **XOR checksum** computation and appending, verified with 100% coverage closure.

RTL Subsystem Functional Verification – SPI, I2C, UART, AHB **|** Bus Protocols, Randomization, SVA
- Validated RTL subsystems (DFFs, FIFOs) and bus protocols (SPI, UART, AHB) using System Verilog testbenches leveraging **constrained randomization**, functional coverage, and **SystemVerilog Assertions** (SVA).